RONISH PRAJAPATI

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EDUCATION

London Metropolitan University / Islington College | *B.Sc (Hons) Computing with AI* 2023 – Present

Campion Academy | *Higher Secondary / A-Levels* 2020 – 2022

EXPERIENCE

Tanvi Tech Pvt. Ltd. | *Machine Learning Intern* Sep 2025 – Dec 2025

- **Built an end-to-end Devanagari OCR pipeline** comparing a Custom CNN model, a pre-trained vision network, and a fine-tuned architecture on the Devanagari dataset to classify characters.
- **Developed the devanagari_ocr_app via FastAPI** to wrap the models, exposing highly performant API endpoints for seamless image character classification.
- **Executed intensive performance benchmarking** and streamlined image data preprocessing workflows to ensure robust evaluation and deployment readiness.

PROJECTS

Palm Mind RAG Engine | *Python, FastAPI, Qdrant, Redis, SQLite, Ollama*

- **Developed an Agentic RAG service** natively using local open-source models (Llama 3.1 & Nomic Embedding), entirely bypassing RAG frameworks like LangChain or LlamaIndex.
- **Implemented a dual-engine data persistence** strategy incorporating a high-speed Redis cluster for multi-turn session tracking and SQLite for reliable transactional metadata storage.
- **Optimized inference pipeline** cold-start latency by designing a **FastAPI lifespan** event hook that automatically preloads and pins local architectures into GPU VRAM on system boot.

Hansei | *Python, PyTorch, Node.js, FastAPI, Qdrant, React Native*

- **Architected and deployed a cross-platform 'clinical companion' app** featuring a React Native mobile frontend, Node.js backend, and a high-performance FastAPI AI engine.
- **Integrated a Retrieval-Augmented Generation (RAG) pipeline** backed by a Qdrant vector database to cross-reference user queries against structured CBT guidelines for reliable chat generation.
- **Designed a Transformer-based deep learning module** utilizing the GoEmotions dataset to support real-time text emotion tracking and automated cognitive reframing.

Deforestation Detection | *Python, PyTorch, ResNet-50, OpenCV, Gradio*

- **Developed an automated land cover classification system** utilizing the EuroSAT dataset to track and monitor deforestation patterns from satellite imagery.
- **Achieved a ~98% test accuracy** with the fine-tuned ResNet-50 model, successfully resolving complex feature classification between forest and pasture terrains.
- **Built a real-time web interface using Gradio** to allow users to upload satellite frames for instant classification and model performance visualization.

GoEmotion Classification | *Python, PyTorch, Transformers, Hugging Face, Sci-Kit Learn*

- **Implemented and benchmarked three deep learning architectures**—Bi-LSTM, Seq2Seq, and an Encoder-only Transformer—to optimize multi-label performance.
- **Achieved a 0.5098 Macro F1-Score** with the Transformer model, successfully outperforming the established Google baseline.

TECHNICAL SKILLS

- **AI & Agentic Frameworks:** RAG-based AI Engines, HuggingFace, Transformers, LLMs, Computer Vision (CNNs, ResNet, OCR), NLP, Qdrant
- **ML & Data Science:** PyTorch, TensorFlow, Sci-Kit Learn, Neural Network Architectures from Scratch, Data Preprocessing
- **Programming & Backend:** Python (Asynchronous), JavaScript (Node.js), FastAPI, Java, Data Structures & Algorithms
- **Infrastructure & Tools:** Docker, Linux, Windows, Secure API Integration, Git, VS Code, Jupyter, Colab

CERTIFICATIONS & REFERENCES

- **Certifications:** Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning (DeepLearning.AI / Stanford).
- **Reference:** Prithivi Maharjan | Lecturer / Tutor at Islington College | prithivi.maharjan18@gmail.com